

Late Breaking Results: Utilization of Hybrid Threshold-Voltage Flip-flops for Power Recovery

Sehyeon Chung^{1,2}, Hyun-chul Hwang³, Byung Su Kim³, Jaeha Lee⁴, Kunhyuk Kang⁵, Taewhan Kim¹

¹*School of Electrical and Computer Engineering, Seoul National University, South Korea*

²*Memory, ³System LSI, ⁴Foundry* Division, Samsung Electronics, South Korea

⁵*Samsung Austin Research Center, USA*

{shchung, tkim}@snucad.snu.ac.kr; {h_c.hwang, bs2014.kim, jaehayo.lee, kunhyuk.kang}@samsung.com

Abstract—As the process technology advances, reducing the leakage power as much as possible is one of the utmost challenging tasks in chip implementation. Utilizing cells with multi-VT (threshold voltage) is known to be a very effective method for optimizing leakage power under timing constraints. However, for the sequential cells on the timing critical paths in circuits, there is no easy way for the multi-VT method to reduce the leakage power unless timing is not sacrificed. To overcome this barrier, we introduce a set of new standard cells called *hybrid-VT flip-flop cells*, each of which is implemented with two different VT types, one implanted onto its master latch while the other onto its slave latch, by which the setup time and clock-to-Q delay can be controlled individually and independently. We confirm that applying our power recovery method utilizing hybrid-VT flip-flop cells to the benchmark circuits, which have already been optimized by the conventional multi-VT cells, is able to further reduce the leakage power by 8.97% with no timing degradation.

Index Terms—threshold voltages, flip-flop, standard cell, low-power design, design-technology co-optimization

I. INTRODUCTION

Multi-VT cells are essential components for implementing low-power and high-performance chips where RVT (regular-VT) cells are mostly used to implement the logic gates on non-critical timing paths to save leakage power while LVT (low-VT) or SLVT (super-low-VT) cells are used to implement some gates on critical timing paths to meet the performance constraint at the expense of leakage power (e.g., [1]). Table I shows the trade-off between delay and leakage power of an inverter, a 2-input NAND, and a flip-flop cell implanted each with RVT, LVT, and SLVT in ASAP7 library [2].

TABLE I

DELAY (ps) AND LEAKAGE POWER (nW) OF REPRESENTATIVE CELLS IN THE ASAP7 LIBRARY [2] IMPLANTED EACH WITH RVT, LVT, AND SLVT.

	Inverter		2-input NAND		Flip-flop	
	delay	power	delay	power	delay [†]	power
RVT	28.3	0.053	45.0	0.106	127.0	0.260
LVT	23.0 (18.7%↓)	0.520 (9.8x↑)	37.8 (16.0%↓)	1.040 (9.8x↑)	96.0 (24.4%↓)	2.554 (9.8x↑)
SLVT	19.3 (31.8%↓)	6.143 (96.5x↑)	32.8 (27.1%↓)	10.281 (96.7x↑)	82.0 (35.4%↓)	25.987 (99.9x↑)

[†]delay = setup time + clock-to-Q delay

Since the delay saving per unit power by VT change on small cells is much bigger than that on large cells, as indicated in Table I, preferentially assigning SLVT to the small cells on the critical timing paths rather than to large cells, such

as flip-flops, is economical in the usage of leakage power. However, to implement modern high-speed circuits beyond 3GHz, we have no choice but to use SLVT flip-flops on critical paths with a very stringent timing budget. Consequently, the circuits will be obliged to dissipate the excessive flip-flop leakage power. To our knowledge, [3] proposed a multi-VT flip-flop. However, the flip-flop is practically infeasible in that it is simply a transistor-level VT assignment and overlooks the rigorous cell layout constraint among the different internal VT implantations. In this work, we propose a new *hybrid-VT flip-flop cells* where different VTs are implanted in the master and slave latches with no design rule violation, allowing independent adjustment of setup time and clock-to-Q delay, thereby presenting a method to recover leakage power.

II. HYBRID THRESHOLD-VOLTAGE FLIP-FLOP CELLS

Concept: Since the starting and ending points of every timing path are defined by the slave latch (i.e., starting from clock-to-Q delay) and master latch (i.e., ending at setup time) of flip-flops, respectively, it will be possible to optimize the leakage power on the flip-flops that are placed on the boundary between the critical and non-critical timing paths if their master and slave latches can be implanted with different VTs. For example, Table II shows the changes in setup time, clock-to-Q delay, and leakage power, experimentally derived, for hybrid-VT flip-flops and a fully SLVT flip-flop in the ASAP7 library [2]. By selectively implanting LVT / SLVT for the master latch and SLVT / LVT for the slave latch, as illustrated in Fig. 1(a) / Fig. 1(b), the leakage power consumed by the fully SLVT flip-flop on a timing path can be reduced while satisfying timing constraints.

TABLE II

DELAY (ps) AND LEAKAGE POWER (nW) OF FLIP-FLOP CELLS IN FIG. 1 IMPLANTED WITH SLVT ONLY, LVT+SLVT, AND SLVT+LVT.

VT (master + slave)	setup time	clock-to-Q delay	leakage power
SLVT + SLVT	8.0	73.9	25.987
LVT + SLVT	16.0 (100.0%↑)	74.3 (0.5%↑)	19.387 (25.4%↓)
SLVT + LVT	8.0 (0.0%)	83.2 (12.6%↑)	16.062 (38.2%↓)

Cell layout: Fig. 2(b) show the layout of DFFHQNx1 cell in ASAP7 library, in which the blue and orange regions correspond to the master and slave latches of the flip-flop

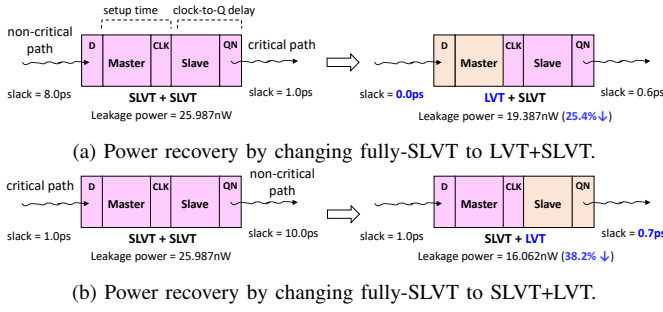


Fig. 1. Leakage power recovery by using hybrid-VT flip-flops in Table II.

circuit in Fig. 2(a), respectively. Figs. 2(c) and (d) show two hybrid-VT implanted layouts derived from the layout in Fig. 2(b), one with LVT for the master latch and SLVT for the slave, the other with SLVT for the master, and LVT for the slave. It should be noted that to meet the design rules in ASAP7 PDK, 2 CPPs (contact poly pitch) spacing is made by inserting two dummy gates in between SLVT and LVT regions.

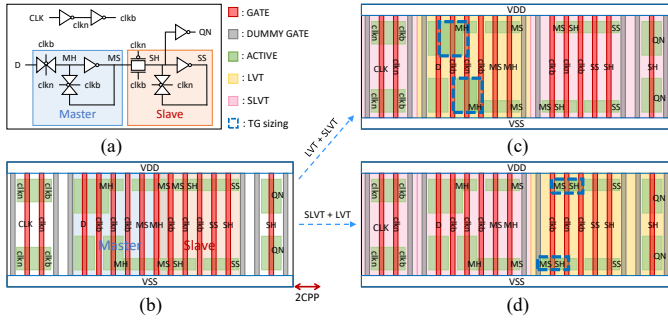


Fig. 2. Flip-flop cell layouts implanted with fully SLVT, hybrid-VT. (a) A flip-flop logic circuit. (b) Fully SLVT implanted layout (DFFHQX1 in [2]) of the flip-flop in (a). (c) LVT+SLVT implanted layout derived from (b). (d) SLVT+LVT implanted layout derived from (b).

Since hybrid-VT flip-flop induces transistor delay difference depending on its VT type, the signal mismatch between MH (transmission gate output in the master latch in Fig. 2) and SH (transmission gate output in the slave latch in Fig. 2) causes synchronization instability. We completely resolve this problem by upsizing the transmission gate of lower VT. The dotted boxes in Figs. 2(c) and (d) indicate the upsized transistors for this purpose.

Power recovery: Fig. 3 illustrates our proposed design flow for power recovery using hybrid-VT flip-flop cells. From the circuit obtained from logic synthesis and place-and-route using state-of-the-art commercial tools, we carry out two additional steps, as indicated by the blue lines in Fig. 3: (1) hybrid-VT flip-flop cell generation, followed by (2) leakage power recovery using hybrid-VT flip-flops.

III. EXPERIMENTAL RESULTS

We evaluated our proposed design flow by implementing it on a Linux machine equipped with 128GB and an AMD Ryzen

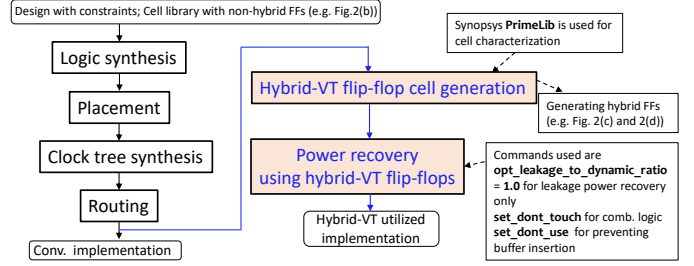


Fig. 3. Design flow for power recovery using hybrid-VT flip-flop cells.

3970X CPU. We used OpenCores benchmark [4] and used Synopsys Design Compiler and Cadence Innovus with ASAP7 PDK [2] to synthesize and implement the circuits. Table III shows a comparison of the total cell area, the worst / total negative slack, and the total leakage power on flip-flops for the circuits optimized by the conventional flow and our proposed flow using hybrid-VT flip-flops. Overall, the proposed flow is able to save leakage power by 8.97% further on average with no timing degradation. As seen in Table III, the very small VT migration (particularly circuit ETHERNET) among the flip-flops results in a big saving on leakage power, which is because about 10× power saving can be made by changing from SLVT (LVT) to LVT (RVT), as revealed in Table I.

TABLE III
AREA, TIMING, AND FLIP-FLOP LEAKAGE POWER OF IMPLEMENTATIONS.

Circuit		Conv. flow	New flow using Hybr-VT FFs	
WB_DMA	Area (μm^2)	6016	6019	0.05%↑
	WNS (ps)	0.000	0.001	met
	TNS (ps)	0.000	0.000	unchanged
	Power (FFs, nW)	0.514	0.489	4.9%↓
	#R/L/SL-VT_FF	484/25/14	484/24/12	
	#Hybrid-VT_FF	0	3 (0.6%)	
AES_CIPHER	Area (μm^2)	23423	23450	0.12%↑
	WNS (ps)	-0.108	-0.107	met
	TNS (ps)	-18.896	-18.860	improved
	Power (FFs, nW)	6.311	6.010	4.8%↓
	#R/L/SL-VT_FF	140/124/266	140/123/238	
	#Hybrid-VT_FF	0	29 (5.5%)	
WB_CONMAX	Area (μm^2)	38199	38204	0.01%↑
	WNS (ps)	-0.008	-0.008	met
	TNS (ps)	-0.058	-0.054	improved
	Power (FFs, nW)	3.313	3.158	4.7%↓
	#R/L/SL-VT_FF	348/399/71	349/399/64	
	#Hybrid-VT_FF	0	6 (0.7%)	
ETHERNET	Area (μm^2)	103020	103099	0.08%↑
	WNS (ps)	-0.191	-0.182	met
	TNS (ps)	-7.701	-6.928	improved
	Power (FFs, nW)	4.393	3.571	18.7%↓
	#R/L/SL-VT_FF	10429/35/79	10439/11/9	
	#Hybrid-VT_FF	0	84 (0.8%)	

Acknowledgement: This work was supported National Research Foundation by Korea Government (2021-R1A2C2008864), Samsung Electronics Co., Ltd (IO201216-08205-01, LSI250306-0010, IO241217-11507-01, IO230102-04421-01), and SAIT (IO250331-12476-01).

REFERENCES

- [1] G. Chua-Eoan, "Methods for optimizing performance and power employing lower threshold voltages," *Patent US8924902B2*, 2010.
- [2] L. T. Clark *et al.*, "Asap7: A 7-nm finfet predictive process design kit," *Microelectronics Journal*, 2016.
- [3] W. Zhang *et al.*, "Low-leakage flip-flops based on dual-threshold and multiple leakage reduction techniques," *JCSC*, 2011.
- [4] Oliscience, "Opencores," 1999. [Online]. Available: <https://opencores.org>